CEE6501 — Lecture 8.1

Member Releases

Plane frames and beams with moment releases (hinges)

Learning Objectives

By the end of this lecture, you will be able to:

- Explain what a **member release** is and why it matters in DSM
- Classify members by **release type** (no hinge / hinge at b / hinge at e / hinges at both ends)
- Derive modified local stiffness relations by enforcing **zero end moment** at a hinge
- Write the resulting **modified local stiffness matrix** and **modified fixed-end force vector**
- Recognize and handle **hinged joints** and why they can cause singular stiffness matrices

Agenda

- **Part 1** — Concepts + notation
- **Part 2** — Derivations (MT = 1, 2, 3)
- **Part 3** — Hinged joints + singularity
- **Part 4** — Wrap-up

Part 1 — Concepts and Notation

What is a member release?

- A **member release** removes a force/moment transfer capability at an end of a member.
- Here we focus on **moment releases** (hinges).
- If an end is hinged, the **end moment must be zero** at that end.

Key constraint (local):

$$M_b = 0 \quad \text{or} \quad M_e = 0$$

Member release types (classification)

Figure placeholder: Fig. 7.1 — Member release types

- **MT = 0:** no hinges (rigid at both ends, normal frame element)
- **MT = 1:** hinge at **beginning** (end **b**)
- **MT = 2:** hinge at **end** (end **e**)
- **MT = 3:** hinges at **both ends**

Local DOFs and end forces (2D frame member)

Local displacement vector (member coordinates):

$$\mathbf{u}^T = \begin{bmatrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{bmatrix} = \begin{bmatrix} u_b & v_b & \theta_b & u_e & v_e & \theta_e \end{bmatrix}$$

Local end-force vector:

$$\mathbf{Q}^T = \begin{bmatrix} Q_1 & Q_2 & Q_3 & Q_4 & Q_5 & Q_6 \end{bmatrix} = \begin{bmatrix} N_b & V_b & M_b & N_e & V_e & M_e \end{bmatrix}$$

Fixed-end force notation (FEFs)

We write local fixed-end forces using the convention:

- Q^F_{xb} = FEF at node **b** in local **x** (axial) → corresponds to **DOF 1**
- Q^F_{yb} = FEF at node **b** in local **y** (shear) → corresponds to **DOF 2**
- Q^F_{mb} = FEF at node **b** (moment) → corresponds to **DOF 3**
- Q^F_{xe} = FEF at node **e** in local **x** (axial) → corresponds to **DOF 4**
- Q^F_{ye} = FEF at node **e** in local **y** (shear) → corresponds to **DOF 5**
- Q^F_{me} = FEF at node **e** (moment) → corresponds to **DOF 6**

So the fixed-end force vector is

$$\mathbf{Q}_f^T = \begin{bmatrix} Q^F_{xb} & Q^F_{yb} & Q^F_{mb} & Q^F_{xe} & Q^F_{ye} & Q^F_{me} \end{bmatrix}$$

Part 2 — Derivations for Released Members

Starting point (MT = 0): standard local stiffness relation

$$\mathbf{Q} = \mathbf{k}\,\mathbf{u} + \mathbf{Q}^F$$

Member releases modify **only**:

- the local stiffness matrix $\mathbf{k}$, and
- the fixed-end force vector $\mathbf{Q}^F$,

so the DSM workflow stays the same.

Base local stiffness relations (no hinges)

Recall from last lecture. For a plane frame member with no hinges, the local end-force components can be written (explicitly) as:

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q^F_{xb} \\
Q_2 &= \frac{EI}{L^3}(12u_2 + 6Lu_3 - 12u_5 + 6Lu_6) + Q^F_{yb} \\
Q_3 &= \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q^F_{mb} \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q^F_{xe} \\
Q_5 &= \frac{EI}{L^3}(-12u_2 - 6Lu_3 + 12u_5 - 6Lu_6) + Q^F_{ye} \\
Q_6 &= \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q^F_{me}
\end{aligned}
$$

Same Equations in Matrix Form

$$\begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \\ Q_6 \end{Bmatrix} = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{Bmatrix} + \begin{Bmatrix} Q_1^F \\ Q_2^F \\ Q_3^F \\ Q_4^F \\ Q_5^F \\ Q_6^F \end{Bmatrix}$$

MT = 0: Matrix Form

$$\mathbf{k}_{\text{frame}} = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix}$$

MT = 1 (hinge at b): enforce $M_b = 0$

Hinge at the beginning means

$$M_b = Q_3 = 0$$

Start from

$$Q_3 = \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F$$

Set $Q_3 = 0$ and solve for u_3 (released rotation at b).

MT = 1: solve for the released rotation u_3

Set $Q_3 = 0$:

$$0 = \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F$$

Multiply by $\frac{L^3}{EI}$ and isolate u_3:

$$4L^2u_3 = -6L(u_2 - u_5) - 2L^2u_6 - \frac{L^3}{EI}Q_{mb}^F$$

$$u_3 = -\frac{3}{2L}(u_2 - u_5) - \frac{1}{2}u_6 - \frac{L}{4EI}Q_{mb}^F$$

Key point: *u_3 is no longer an independent DOF in the member relations. The row/column corresponding to this DOF can be expressed in terms of a linear combination of other rows/columns in the matrix.*

MT = 1: substitute u_3 back into Q_2, Q_5, Q_6

Resulting modified equations (MT = 1):

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= \frac{EI}{L^3}(3u_2 - 3u_5 + 3Lu_6) + \left(Q_{yb}^F - \frac{3}{2L}Q_{mb}^F\right) \\
Q_3 &= 0 \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= \frac{EI}{L^3}(-3u_2 + 3u_5 - 3Lu_6) + \left(Q_{ye}^F + \frac{3}{2L}Q_{mb}^F\right) \\
Q_6 &= \frac{EI}{L^3}\left(3Lu_2 - 3Lu_5 + 3L^2u_6\right) + \left(Q_{me}^F - \frac{1}{2}Q_{mb}^F\right)
\end{aligned}
$$

MT = 1: stiffness matrix

row/column 3 are 0's:

$$\mathbf{k}_{(MT=1)} = \frac{EI}{L^3}\begin{bmatrix} \frac{AL^2}{I} & 0 & \boxed{0} & -\frac{AL^2}{I} & 0 & 0 \\ 0 & 3 & \boxed{0} & 0 & -3 & 3L \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \\ -\frac{AL^2}{I} & 0 & \boxed{0} & \frac{AL^2}{I} & 0 & 0 \\ 0 & -3 & \boxed{0} & 0 & 3 & -3L \\ 0 & 3L & \boxed{0} & 0 & -3L & 3L^2 \end{bmatrix}$$

MT = 1: fixed end force vector

row/column 3 are 0's:

$$\mathbf{Q}^F_{(MT=1)} = \begin{bmatrix} F^F_{xb} \\ F^F_{yb} - \frac{3}{2L} F^F_{mb} \\ \boxed{0} \\ F^F_{xe} \\ F^F_{ye} + \frac{3}{2L} F^F_{mb} \\ F^F_{me} - \frac{1}{2} F^F_{mb} \end{bmatrix}$$

MT = 2 (hinge at e): enforce $M_e = 0$

Hinge at the end means

$$M_e = Q_6 = 0$$

Start from

$$Q_6 = \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F$$

Set $Q_6 = 0$ and solve for u_6 (released rotation at e).

MT = 2: solve for the released rotation u_6

Set $Q_6 = 0$:

$$0 = \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F$$

Multiply by $\frac{L^3}{EI}$ and isolate u_6:

$$4L^2u_6 = -6L(u_2 - u_5) - 2L^2u_3 - \frac{L^3}{EI}Q_{me}^F$$

$$u_6 = -\frac{3}{2L}(u_2 - u_5) - \frac{1}{2}u_3 - \frac{L}{4EI}Q_{me}^F$$

MT = 2: substitute u_6 back into Q_2, Q_3, Q_5

Resulting modified equations (MT = 2):

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= \frac{EI}{L^3}(3u_2 + 3Lu_3 - 3u_5) + \left(Q_{yb}^F - \frac{3}{2L}Q_{me}^F\right) \\
Q_3 &= \frac{EI}{L^3}\left(3Lu_2 + 3L^2u_3 - 3Lu_5\right) + \left(Q_{mb}^F - \frac{1}{2}Q_{me}^F\right) \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= \frac{EI}{L^3}(-3u_2 - 3Lu_3 + 3u_5) + \left(Q_{ye}^F + \frac{3}{2L}Q_{me}^F\right) \\
Q_6 &= 0
\end{aligned}
$$

MT = 2: stiffness matrix

Row and column 6 are zero:

$$\mathbf{k}_{(MT=2)} = \frac{EI}{L^3}\begin{bmatrix} \frac{AL^2}{I} & 0 & 0 & -\frac{AL^2}{I} & 0 & \boxed{0} \\ 0 & 3 & 3L & 0 & -3 & \boxed{0} \\ 0 & 3L & 3L^2 & 0 & -3L & \boxed{0} \\ -\frac{AL^2}{I} & 0 & 0 & \frac{AL^2}{I} & 0 & \boxed{0} \\ 0 & -3 & -3L & 0 & 3 & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \end{bmatrix}$$

MT = 2: fixed-end force vector

Row/column 6 are zero:

$$\mathbf{Q}_{f,(MT=2)} = \begin{bmatrix} Q_{xb}^F \\ Q_{yb}^F - \dfrac{3}{2L} Q_{me}^F \\ Q_{mb}^F - \dfrac{1}{2} Q_{me}^F \\ Q_{xe}^F \\ Q_{ye}^F + \dfrac{3}{2L} Q_{me}^F \\ \boxed{0} \end{bmatrix}$$

MT = 3 (hinges at both ends): enforce $M_b = 0$ and $M_e = 0$

Hinges at both ends mean

$$Q_3 = 0, \qquad Q_6 = 0$$

We now have **two constraints** that couple u_3 and u_6.

Start from the base moment equations:

$$Q_3 = \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F$$

$$Q_6 = \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F$$

MT = 3: solve the coupled system for u_3 and u_6

Set $Q_3 = 0$ and $Q_6 = 0$, multiply both by $\frac{L^3}{EI}$:

$$0 = 6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6 + \frac{L^3}{EI}Q^F_{mb}$$

$$0 = 6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6 + \frac{L^3}{EI}Q^F_{me}$$

Solving gives (one convenient form):

$$u_3 = -\frac{1}{L}(u_2 - u_5) - \frac{L}{6EI}(2Q^F_{mb} - Q^F_{me})$$

$$u_6 = -\frac{1}{L}(u_2 - u_5) - \frac{L}{6EI}(2Q^F_{me} - Q^F_{mb})$$

MT = 3: substitute u_3 and u_6 back Q_2, Q_5

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= Q_{yb}^F - \frac{1}{L}(Q_{mb}^F + Q_{me}^F) \\
Q_3 &= 0 \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= Q_{ye}^F + \frac{1}{L}(Q_{mb}^F + Q_{me}^F) \\
Q_6 &= 0
\end{aligned}
$$

Interpretation

- Notice that Q_2 **and** Q_5 **no longer contain any displacement terms**.
- They are **purely functions of the fixed-end forces**.
- The bending stiffness contributions have completely disappeared.
- This means the element **cannot develop bending resistance**.
- The transverse shear forces are determined solely by equilibrium of the fixed-end moments.
- The element now behaves as a **pure axial (truss-type) member**.

MT = 3: stiffness matrix

Not only are row/column 3 and 6 are zero, but also row/column 2 and 5:

$$\mathbf{k}_{(MT=3)} = \frac{EA}{L}\begin{bmatrix} 1 & 0 & \boxed{0} & -1 & 0 & \boxed{0} \\ 0 & 0 & \boxed{0} & 0 & 0 & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \\ -1 & 0 & \boxed{0} & 1 & 0 & \boxed{0} \\ 0 & 0 & \boxed{0} & 0 & 0 & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \end{bmatrix}$$

What does this stiffness matrix look like?

MT = 3: fixed-end force vector

Row/column 3 and 6 are zero:

$$\mathbf{Q}^F_{(MT=3)} = \begin{bmatrix} Q^F_{xb} \\ Q^F_{yb} - \frac{1}{L}(Q^F_{mb} + Q^F_{me}) \\ \boxed{0} \\ Q^F_{xe} \\ Q^F_{ye} + \frac{1}{L}(F^F_{mb} + Q^F_{me}) \\ \boxed{0} \end{bmatrix}$$

Implementation note: keep 6×6, but released DOFs have zero rows/cols

- Even though releases reduce the **independent** member coordinates, we often keep a **6×6** member form.
- Released moment rows/cols (and their corresponding fixed-end terms) become **zero** in the stiffness matrix and FEF vector.
- This keeps the DSM assembly code clean: same DOF mapping size for every member.

Then transform each element to global coordinates as usual:

$$\mathbf{K} = \mathbf{T}^T \mathbf{k} \mathbf{T}, \qquad \mathbf{F}^F = \mathbf{T}^T \mathbf{Q}^F$$

Part 3 — Hinged Joints and Matrix Singularity

Hinged joints in beams and plane frames

A **hinged joint** is a joint where all members meeting at that joint are connected by **moment releases**.

Figure placeholder: Fig. 7.2 — Example plane frame showing a hinged joint

Physical consequence:

- The joint **cannot transmit bending moment**.
- The joint is **free to rotate**.
- Axial and shear forces may still be transmitted.

What does this mean for the DOFs?

At a hinged joint:

- The rotational DOF θ_j is not constrained (it is free).
- But no connected member provides **rotational stiffness**.
- The joint rotation does not generate bending moment.

In other words: The structure has a rotation that costs **zero strain energy**.

DSM partitioned equilibrium

Recall the partitioned system:

$$\begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fr} \\ \mathbf{K}_{rf} & \mathbf{K}_{rr} \end{bmatrix} \begin{bmatrix} \mathbf{u}_f \\ \mathbf{u}_r \end{bmatrix} = \begin{bmatrix} \mathbf{F}_f \\ \mathbf{F}_r \end{bmatrix}$$

We solve for the unknown displacements from

$$\mathbf{K}_{ff}\mathbf{u}_f = \mathbf{F}_f - \mathbf{K}_{fr}\mathbf{u}_r$$

This requires that $\mathbf{K}_{ff}$ be **invertible**.

What happens if θ_j is left in the free set?

The rotational DOF θ_j is not restrained, so it is placed in the free displacement vector $\mathbf{u}_f$.

As a free DOF, it contributes a corresponding row and column to the free stiffness matrix $\mathbf{K}_{ff}$.

However, at a fully hinged joint, no connected element provides rotational stiffness. From our earlier derivations, the entire row and column associated with θ_j are zero.

Therefore:

- $\mathbf{K}_{ff}$ contains a **zero row and column**.
- $\mathbf{K}_{ff}$ becomes **singular** (non-invertible).
- The system of equations does not admit a unique displacement solution.

Important clarification

This situation does **not** indicate a physical instability.

The structure may be completely stable in translation and capable of carrying load.

The singularity arises because a rotational DOF was left in the free set **without any associated stiffness**.

In other words:

- The joint can rotate freely.
- That rotation does not generate internal moment.
- The stiffness matrix has no resistance in that direction.

The issue is therefore **mathematical**, not structural.

To obtain a well-posed system of equations, this DOF must be treated properly in the DSM formulation.

Two Modeling Approaches

Two practical ways to avoid singularity:

Method A — Artificially restrain (fix) the joint rotation

- Put the hinged-joint rotation into the **restrained set** $\mathbf{u}_r$ (not in $\mathbf{u}_f$)
- The computed reaction moment at that DOF ends up being **zero** anyway

Method B — Do not treat the joint as hinged (in the global DOF sense)

- Model the joint as rigidly connected to **one** of the members meeting there
- Assign the joint three DOFs (u_x, u_y, θ) as usual
- Other members connect by hinges (moment releases) so the net joint moment remains zero

Part 4 - Method A, Fictitious Fixity

The Main Idea

The most straightforward remedy for the singularity is:

> *Eliminate the rotational DOF of hinged joints by modeling such joints as restrained (fixed) against rotation.*

This is imposing **fictitious fixity** that does not change global structural behavior.

Internal Moment at a Hinged Joint

At a fully hinged joint, j, by formulation:

- All connected members have moment releases.
- The joint **cannot transmit bending moment**.
- No internal bending moment can develop at that node.

Therefore,

$$M_j = 0$$

This is a structural fact, not something we impose.

Artificially Fixing the Rotation

Suppose we impose fictitious fixity and set

$$\theta_j = 0$$

instead of leaving θ_j in the free displacement vector.

What Happens to the Calculated Reactions at that DOF?

Recall: After solving for the free DOFs, u_f, we compute the reactions, F_r, at all the restrained DOFs.

For the fictitiously fixed rotation, the computed reaction will be (you can verify this):

$$F_{m,j} = 0$$

This confirms that, even if fixed, the hinge does not develop moment.

Thus, the artificial restraint does not introduce spurious forces.

Why the Structural Response is Unchanged

Imposing $\theta_j = 0$ does not change the moments developed in the joint, because:

- No connected element provides rotational stiffness.
- The global stiffness matrix contains no coupling terms involving θ_j.
- The equilibrium equations do not depend on that rotational DOF.

Because no bending stiffness exists at the joint, fixing the rotation does not generate any additional reaction moment (moment is related to displacement through stiffness).

You can think of DOF j as a restrained DOF that simply develops **no reaction load**.

Implementing with DSM Partitioning

Instead of leaving θ_j in the free displacement vector $\mathbf{u}_f$,

we classify it as a **restrained DOF**.

So:

- It is placed in $\mathbf{u}_r$.
- It is removed from $\mathbf{K}_{ff}$.
- It is no longer part of the system we invert:

$$\mathbf{K}_{ff}^{-1}$$

The zero-stiffness row and column disappear from the matrix being solved.

Method A Summary

Method A is therefore a **numerical bookkeeping device**:

- It removes a zero-stiffness DOF from the unknown set,
- Eliminates the singularity in $\mathbf{K}_{ff}$,
- Preserves the correct internal force distribution.

The physics is unchanged but the algebra becomes well-posed.

Method A — Step-by-Step Procedure

1. **Identify the hinged joint**

    - All connected member ends at that node are moment-released.
    - The joint cannot transmit bending moment.

2. **Keep the rotational DOF in the global numbering**

    - The node still has (u_x, u_y, θ).
    - We do not remove the DOF from the system definition.

3. **Classify the rotation as a restrained DOF**

    - Place θ_j in the restrained displacement vector $\mathbf{u}_r$.
    - Impose the artificial condition

$$\theta_j = 0$$

This removes the hinge rotation from the unknown set, u_f.

4. **Assemble the global stiffness matrix normally**

    - Because all connected elements are released, the assembled stiffness terms associated with θ_j are zero.
    - No rotational stiffness is introduced.

5. **Solve for the free DOFs**

    - θ_j is not included in $\mathbf{u}_f$.
    - The matrix $\mathbf{K}_{ff}$ no longer contains a zero row/column.
    - The system becomes invertible.

6. **Compute reactions**

    Reactions are obtained from

$$\mathbf{F}_r = \mathbf{K}_{rf}\mathbf{u}_f + \mathbf{K}_{rr}\mathbf{u}_r + \mathbf{F}_r^F$$

    The reaction moment at the fictitiously fixed DOF evaluates to

$$M_j \approx 0$$

    confirming hinge behavior.

Method B — Model the joint as rigidly connected to one member

Alternative idea from the text:

- Treat the joint as rigidly connected to **one (and only one)** of the members meeting at the joint.
- Other members meeting there are connected by **hinges**.

Conceptual justification:

- No external moment is applied at the hinged joint.
- Therefore, the moments at the ends of all members meeting at the joint must sum to zero.
- With hinged connections, those member-end moments at the joint are zero anyway.

So this approach avoids a singular global system while remaining statically consistent.

Method B — Step-by-step

1. **Identify** the hinged joint and list connected members.

2. **Pick one "reference" member** at the joint

- This member is modeled as **rigidly connected** to the joint.
- The joint rotation DOF is associated with this member connection.

3. **Model all other members as hinged at the joint**

- Use released-end formulations (MT=1 or MT=2) at that joint end.
- Their end moments at the joint are enforced as zero.

4. **Treat the joint as a normal rigid node in global DOF bookkeeping**

- Keep the joint DOFs (u_x, u_y, θ).
- Now θ can remain a FREE DOF without producing an all-zero row/column in $\mathbf{K}_{ff}$.

5. **Assemble and solve**

- The presence of the one rigid connection avoids the purely unsupported rotation mechanism.
- The hinge behavior is still represented through releases on the other members.

Method A vs Method B (clear takeaway)

The root cause of the singularity is:

> *leaving a fully-hinged joint rotation as a **FREE DOF** inside $\mathbf{K}_{ff}$, even though it has **zero stiffness**.*

Method A: keep the rotation DOF but classify it as **RESTRAINED** (artificial fixity)

- Avoids a zero-stiffness FREE DOF in $\mathbf{K}_{ff}$
- Reaction moment at that DOF comes out **0**

Method B: keep the joint as a normal 3-DOF node, but make it rigid to exactly **one** member

- Avoids the unsupported mechanism DOF
- Other members meet by hinges so end moments at the joint are still zero

Part 5 — Implementing Method A

Problem data (from the example)

Assume member properties are constant for all members:

- $E = 200 \times 10^6 \text{ kN/m}^2 \ (= 200 \text{ GPa})$
- $A = 0.0065 \text{ m}^2 \ (= 6500 \text{ mm}^2)$
- $I = 150 \times 10^{-6} \text{ m}^4 (= 150 \times 10^6 \text{ mm}^4)$
- $L = 5 \text{ m}$

Local DOF order for each 2D frame member:

$$[u_{xb},\ u_{yb},\ \theta_b,\ u_{xe},\ u_{xe},\ \theta_e]$$

Fixed-end force notation:

$$\begin{aligned} \text{Node } b: &\quad Q^F_{xb},\ Q^F_{yb},\ Q^F_{mb} \\ \text{Node } e: &\quad Q^F_{xe},\ Q^F_{ye},\ Q^F_{me} \end{aligned}$$

Member 1 (MT = 2)

Member 1 has a **hinge at end** e, so it is **MT = 2**.

Local stiffness matrix (MT = 2):

$$\mathbf{k}^{(1)} = \begin{bmatrix} 260{,}000 & 0 & 0 & -260{,}000 & 0 & 0 \\ 0 & 720 & 3{,}600 & 0 & -720 & 0 \\ 0 & 3{,}600 & 18{,}000 & 0 & -3{,}600 & 0 \\ -260{,}000 & 0 & 0 & 260{,}000 & 0 & 0 \\ 0 & -720 & -3{,}600 & 0 & 720 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Local fixed-end forces for uniform transverse load $w = 19.2\ \mathrm{kN/m}$:

First, fully-fixed (rigid) FEF component.

No inclined force or member, axial = 0:

$$Q_{xb}^F = Q_{xe}^F = 0$$

Shear and moment:

$$Q_{yb}^F = Q_{ye}^F = \frac{wL}{2} = \frac{19.2(5)}{2} = 48\ \mathrm{kN}$$

$$Q_{mb}^F = -Q_{me}^F = \frac{wL^2}{12} = \frac{19.2(5)^2}{12} = 40\ \mathrm{kN\text{-}m}$$

Then modify for MT = 2 (hinge at end e):

$$\mathbf{Q}^{F(1)} = \begin{bmatrix} Q^F_{xb} \\ Q^F_{yb} - \frac{3}{2L} F^F_{me} \\ Q^F_{mb} - \frac{1}{2} F^F_{me} \\ Q^F_{xe} \\ Q^F_{ye} + \frac{3}{2L} F^F_{me} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 60 \\ 60 \\ 0 \\ 36 \\ 0 \end{bmatrix}$$

Transformation matrix: member 1 is vertical ($\theta = 90^\circ$, $c = 0$, $s = 1$)

$$\mathbf{T}^{(1)} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Local → Global Map

$$[1, 2, 3, 4, 5, 6]_{\text{local}} = [1, 2, 3, 4, 5, 6]_{\text{global}}$$

Member 2 (MT = 1)

Member 2 has a **hinge at end** b, so it is **MT = 1**.

Local stiffness matrix (MT = 1):

$$\mathbf{k}^{(2)} = \begin{bmatrix} 260{,}000 & 0 & 0 & -260{,}000 & 0 & 0 \\ 0 & 720 & 0 & 0 & -720 & 3{,}600 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -260{,}000 & 0 & 0 & 260{,}000 & 0 & 0 \\ 0 & -720 & 0 & 0 & 720 & -3{,}600 \\ 0 & 3{,}600 & 0 & 0 & -3{,}600 & 18{,}000 \end{bmatrix}$$

Local fixed-end forces (point load case shown in the example):

Rigid-member (fully fixed) FEF components used in the example:

$$Q_{xb}^F = Q_{xe}^F = 0$$

$$Q_{yb}^F = Q_{ye}^F = 150\text{kN}$$

$$Q_{mb}^F = -Q_{me}^F = 187.5\text{kN-m}$$

Modify for MT = 1 (hinge at end b):

$$\mathbf{Q}^{F(2)} = \begin{bmatrix} Q^F_{xb} \\ Q^F_{yb} - \frac{3}{2L} Q^F_{mb} \\ 0 \\ Q^F_{xe} \\ Q^F_{ye} + \frac{3}{2L} Q^F_{mb} \\ Q^F_{me} - \frac{1}{2} Q^F_{mb} \end{bmatrix} = \begin{bmatrix} 0 \\ 93.75 \\ 0 \\ 0 \\ 206.25 \\ -281.25 \end{bmatrix}$$

Transformation matrix: member 2 is horizontal (aligned with global), so

$$\mathbf{T}^{(2)} = \mathbf{I}_{6\times 6}$$

$$\mathbf{T}^{(2)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Local → Global Map

$$[1, 2, 3, 4, 5, 6]_{\text{local}} = [2, 3, 4, 5, 6, 7]_{\text{global}}$$

Member 3 (MT = 1)

At the start of member 3 there is a **pin support**.

We have two equivalent modeling choices:

Option 1 — Treat as standard frame element (MT = 0)

- Use the full 6×6 local stiffness matrix (no releases).
- Enforce $u_x = 0$ and $u_y = 0$ in the restrained partition.
- Leave the rotational DOF free.
- This produces the **same final structural response**.

Option 2 — Model the pin as a member release (MT = 1)

- A pin restrains translations but **cannot transmit moment**.
- From an element standpoint, this is equivalent to a hinge at the beginning node (b).
- Therefore member 3 can be modeled as **MT = 1**.

In this example, Option 2 is convenient because:

- Geometrically, member 3 matches member 2.
- Same release condition (hinge at b).
- So we can reuse:

$$\mathbf{k}^{(3)} = \mathbf{k}^{(2)}$$

- No member loading:

$$\mathbf{Q}_f^{(3)} = \mathbf{0}$$

- Vertical member:

$$\theta = 90^\circ \quad (c = 0,\ s = 1)$$

- Mapping

$$[1, 2, 3, 4, 5, 6]_{\text{local}} = [10, 11, 12, 7, 8, 9]_{\text{global}}$$

Note: Both approaches are correct — using MT = 1 simply keeps the setup cleaner.

Try this process with a standard 6x6 element (in class excercise)

In [2093…

```
import numpy as np
import pandas as pd

# -------------------------
# Problem data (hard-coded)
# -------------------------
E = 200e6           # kPa
A = 0.0065          # m^2
I = 150e-6          # m^4
L = 5.0             # m
```

In [2094…

```
# -------------------------
# Member 1 (MT=2)
# -------------------------
k1 = np.array([
    [260000.,     0.,     0., -260000.,     0.,     0.],
    [    0.,   720.,  3600.,      0.,  -720.,     0.],
    [    0.,  3600., 18000.,      0., -3600.,     0.],
    [-260000.,    0.,     0.,  260000.,    0.,     0.],
    [    0.,  -720., -3600.,     0.,   720.,     0.],
    [    0.,     0.,     0.,      0.,     0.,     0.]
], dtype=float)

# Local fixed-end vector for member 1 (given by the example result)
Qf1 = np.array([0., 60., 60., 0., 36., 0.], dtype=float)

# Transformation matrix for theta=90deg -> c=0, s=1
T1 = np.array([
    [ 0.,  1., 0., 0., 0., 0.],
    [-1.,  0., 0., 0., 0., 0.],
    [ 0.,  0., 1., 0., 0., 0.],
    [ 0.,  0., 0., 0., 1., 0.],
    [ 0.,  0., 0.,-1., 0., 0.],
    [ 0.,  0., 0., 0., 0., 1.]
], dtype=float)

# Mapping
m1 = np.array([1,2,3,4,5,6])
```

In [2095…

```
# -------------------------
# Member 2 (MT=1)
# -------------------------
k2 = np.array([
    [260000.,      0.,      0., -260000.,      0.,      0.],
    [     0.,    720.,      0.,       0.,   -720.,   3600.],
    [     0.,      0.,      0.,       0.,      0.,      0.],
    [-260000.,     0.,      0.,  260000.,      0.,      0.],
    [     0.,   -720.,      0.,       0.,    720., -3600.],
    [     0.,   3600.,      0.,       0., -3600., 18000.]
], dtype=float)

# Local fixed-end vector for member 2 (example result)
Qf2 = np.array([0., 93.75, 0., 0., 206.25, -281.25], dtype=float)

# Horizontal member -> no transformation
T2 = np.eye(6, dtype=float)

# Mapping
m2 = np.array([4,5,6,7,8,9])
```

In [2096…

```
# -------------------------
# Member 3 (MT=1)
# -------------------------
k3 = k2.copy()
Qf3 = np.zeros(6, dtype=float)
T3 = T1.copy()
m3 = np.array([10,11,12,7,8,9])
```

In [2097…

```
k_list = [k1, k2, k3]
T_list = [T1, T2, T3]
Qf_list = [Qf1, Qf2, Qf3]
map_list = [m1, m2, m3]   # 1-based DOF maps

# Global system size (still using 1-based maps here)
ndof = int(np.max(np.concatenate(map_list)))
```

In [2098…

```
## USER SPECIFIED

# Applied Load
F_global = np.zeros(ndof, dtype=float)
F_global[4-1] = 90.75

# Restrained DOFs
dof_restrained_1based = np.array([1, 2, 3, 10, 11], dtype=int)

# Fictitious restrained DOFs
dof_fictitious_1based = np.array([6, 12], dtype=int)

# Combine and sort
dof_restrained_1based = np.sort(
    np.concatenate((dof_restrained_1based, dof_fictitious_1based))
)

print(dof_restrained_1based)
```

```
[ 1  2  3  6 10 11 12]
```

In [2099…

```
K_global = np.zeros((ndof, ndof), dtype=float)
F_fef_global = np.zeros(ndof, dtype=float)

for i in range(3):

    k_local = k_list[i]
    T = T_list[i]
    Qf_local = Qf_list[i]
    dof_map = map_list[i]   # still 1-based

    # Transform to global
    K = T.T @ k_local @ T
    F_fef = T.T @ Qf_local

    # Scatter-add (convert to 0-based inside loop)
    for a in range(6):
        A = dof_map[a] - 1

        F_fef_global[A] += F_fef[a]

        for b in range(6):
            B = dof_map[b] - 1
            K_global[A, B] += K[a, b]
```

In [2100…

```
print_matrix_scaled(K_global)
```

```
K = 1e+03 ×
1 | 0.7 0.0 -3.6 -0.7 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
2 | 0.0 260.0 0.0 0.0 -260.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
3 | -3.6 0.0 18.0 3.6 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
4 | -0.7 0.0 3.6 260.7 0.0 0.0 -260.0 0.0 0.0 0.0 0.0 0.0
5 | 0.0 -260.0 0.0 0.0 260.7 0.0 0.0 -0.7 3.6 0.0 0.0 0.0
6 | 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
7 | 0.0 0.0 0.0 -260.0 0.0 0.0 260.7 0.0 3.6 -0.7 0.0 0.0
8 | 0.0 0.0 0.0 0.0 -0.7 0.0 0.0 260.7 -3.6 0.0 -260.0 0.0
9 | 0.0 0.0 0.0 0.0 3.6 0.0 3.6 -3.6 36.0 -3.6 0.0 0.0
10 | 0.0 0.0 0.0 0.0 0.0 0.0 -0.7 0.0 -3.6 0.7 0.0 0.0
11 | 0.0 0.0 0.0 0.0 0.0 0.0 0.0 -260.0 0.0 0.0 260.0 0.0
12 | 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
```

In [2101…

```
(
    K_ff,
    K_fr,
    K_rf,
    K_rr,
    f_f,
    f_r,
    f_fef_f,
    f_fef_r,
    free_dofs,
    restrained_dofs,
) = partition_system(
    K_global,
    F_global,
    F_fef_global,
    dof_restrained_1based
)
```

$$u_f = K_{ff}^{-1}(F_f - F_f^F - K_{fr}u_r)$$

$$F_r = K_{rf}u_f + K_{rr}u_r + F_r^F$$

In [2102…

```
rhs = f_f - f_fef_f
u_f = np.linalg.solve(K_ff, rhs)

F_r = K_rf @ u_f + f_fef_r

u_global = assemble_global_displacements(u_f, free_dofs, restrained_dofs
f_global_complete = assemble_global_forces(f_f, F_r, free_dofs, restrair
```

In [2103…

```
print_dsm_results(
    u_global,
    f_global_complete,
    dof_restrained_1based,
    dof_fictitious_1based,
    disp_in_mm=True
)
```

```
  DOF  Type      Status  Disp (mm / rad)  Load (kN / kN·m)
    1   u_x       Fixed            0.000          -125.918
    2   u_y       Fixed            0.000            89.168
    3 theta       Fixed            0.000           389.592
    4   u_x        Free           91.553            90.750
    5   u_y        Free           -0.343             0.000
    6 theta Fictitious            0.000             0.000
    7   u_x        Free           91.319             0.000
    8   u_y        Free           -0.811             0.000
    9 theta        Free           -0.001             0.000
   10   u_x       Fixed            0.000           -60.832
   11   u_y       Fixed            0.000           210.832
   12 theta Fictitious            0.000             0.000
```

In [2104…

```
# Element 1,2,3
print_element(1, u_global, m1, T1, k1, Qf1)
print_element(2, u_global, m2, T2, k2, Qf2)
print_element(3, u_global, m3, T3, k3, Qf3)
```

```
Element 1
------------------------------------------------------------
u (global) [m&rad] : [ 0.     0.     0.     0.092 -0.     0.   ]
v (local)  [m&rad] : [ 0.     0.     0.    -0.    -0.092  0.   ]
q (local)  [kN&kN·m]    : [ 89.168 125.918 389.592 -89.168 -29.9
18   0.   ]

Element 2
------------------------------------------------------------
u (global) [m&rad] : [ 0.092 -0.     0.     0.091 -0.001 -0.001]
v (local)  [m&rad] : [ 0.092 -0.     0.     0.091 -0.001 -0.001]
q (local)  [kN&kN·m]    : [  60.832   89.168    0.     -60.832
210.832 -304.158]

Element 3
------------------------------------------------------------
u (global) [m&rad] : [ 0.     0.     0.     0.091 -0.001 -0.001]
v (local)  [m&rad] : [ 0.     0.     0.    -0.001 -0.091 -0.001]
q (local)  [kN&kN·m]    : [ 210.832   60.832    0.    -210.832
-60.832  304.158]
```

Part 6 — Implementing Method B

In [2105…

```
import numpy as np
import pandas as pd

# -------------------------
# Problem data (hard-coded)
# -------------------------
E = 200e6           # kPa
A = 0.0065          # m^2
I = 150e-6          # m^4
L = 5.0             # m
```

In [2106…

```
# -------------------------
# Member 1 (MT=2)
# -------------------------
k1 = np.array([
    [260000.,     0.,     0., -260000.,     0.,     0.],
    [    0.,   720.,  3600.,      0.,  -720.,     0.],
    [    0.,  3600., 18000.,      0., -3600.,     0.],
    [-260000.,    0.,     0.,  260000.,    0.,     0.],
    [    0.,  -720., -3600.,     0.,   720.,     0.],
    [    0.,     0.,     0.,      0.,     0.,     0.]
], dtype=float)

# Local fixed-end vector for member 1 (given by the example result)
Qf1 = np.array([0., 60., 60., 0., 36., 0.], dtype=float)

# Transformation matrix for theta=90deg -> c=0, s=1
T1 = np.array([
    [ 0.,  1., 0., 0., 0., 0.],
    [-1.,  0., 0., 0., 0., 0.],
    [ 0.,  0., 1., 0., 0., 0.],
    [ 0.,  0., 0., 0., 1., 0.],
    [ 0.,  0., 0.,-1., 0., 0.],
    [ 0.,  0., 0., 0., 0., 1.]
], dtype=float)

# Mapping
m1 = np.array([1,2,3,4,5,6])
```

In [2107…

```
# -------------------------
# Member 2 (MT=0)
# -------------------------

factor = (E * I) / (L**3)

k2 = factor * np.array(
    [
        [ (A*L**2)/I,  0.0,       0.0,      -(A*L**2)/I,  0.0,       0.0],
        [ 0.0,         12.0,      6.0*L,     0.0,        -12.0,      6.0*L]
        [ 0.0,         6.0*L,     4.0*L**2,  0.0,        -6.0*L,     2.0*L*
        [-(A*L**2)/I,  0.0,       0.0,       (A*L**2)/I,  0.0,       0.0],
        [ 0.0,        -12.0,     -6.0*L,     0.0,         12.0,     -6.0*L]
        [ 0.0,         6.0*L,     2.0*L**2,  0.0,        -6.0*L,     4.0*L*
    ],
    dtype=float,
)

# Local fixed-end force vector (MT=0, no releases)
Qf2 = np.array([0.0, 150.0, 187.5, 0.0, 150.0, -187.5], dtype=float)

# Horizontal member -> no transformation
T2 = np.eye(6, dtype=float)

# Mapping
m2 = np.array([4,5,6,7,8,9])
```

In [2108…

```
# -------------------------
# Member 3 (MT=0)
# -------------------------
k3 = k2.copy()

Qf3 = np.zeros(6, dtype=float)
T3 = T1.copy()
m3 = np.array([10,11,12,7,8,9])
```

In [2109…

```
k_list = [k1, k2, k3]
T_list = [T1, T2, T3]
Qf_list = [Qf1, Qf2, Qf3]
map_list = [m1, m2, m3]   # 1-based DOF maps

# Global system size (still using 1-based maps here)
ndof = int(np.max(np.concatenate(map_list)))
```

In [2110…

```
## USER SPECIFIED

# Applied Load
F_global = np.zeros(ndof, dtype=float)
F_global[4-1] = 90.75

# Restrained DOFs
dof_restrained_1based = np.array([1, 2, 3, 10, 11], dtype=int)

# Fictitious restrained DOFs
dof_fictitious_1based = np.array([], dtype=int)

# Combine and sort
dof_restrained_1based = np.sort(
    np.concatenate((dof_restrained_1based, dof_fictitious_1based))
)
```

In [2111…

```
K_global = np.zeros((ndof, ndof), dtype=float)
F_fef_global = np.zeros(ndof, dtype=float)

for i in range(3):

    k_local = k_list[i]
    T = T_list[i]
    Qf_local = Qf_list[i]
    dof_map = map_list[i]   # still 1-based

    # Transform to global
    K = T.T @ k_local @ T
    F_fef = T.T @ Qf_local

    # Scatter-add (convert to 0-based inside loop)
    for a in range(6):
        A = dof_map[a] - 1

        F_fef_global[A] += F_fef[a]

        for b in range(6):
            B = dof_map[b] - 1
            K_global[A, B] += K[a, b]
```

In [2112…

```
(
    K_ff,
    K_fr,
    K_rf,
    K_rr,
    f_f,
    f_r,
    f_fef_f,
    f_fef_r,
    free_dofs,
    restrained_dofs,
) = partition_system(
    K_global,
    F_global,
    F_fef_global,
    dof_restrained_1based
)
```

In [2113…

```
rhs = f_f - f_fef_f
u_f = np.linalg.solve(K_ff, rhs)

F_r = K_rf @ u_f + f_fef_r

u_global = assemble_global_displacements(u_f, free_dofs, restrained_dofs
f_global_complete = assemble_global_forces(f_f, F_r, free_dofs, restrair
```

In [2114…

```
print_dsm_results(
    u_global,
    f_global_complete,
    dof_restrained_1based,
    dof_fictitious_1based,
    disp_in_mm=True
)
```

```
 DOF  Type Status  Disp (mm / rad)  Load (kN / kN·m)
   1   u_x  Fixed            0.000          -125.918
   2   u_y  Fixed            0.000            89.168
   3 theta  Fixed            0.000           389.592
   4   u_x   Free           91.553            90.750
   5   u_y   Free           -0.343             0.000
   6 theta   Free           -0.007             0.000
   7   u_x   Free           91.319             0.000
   8   u_y   Free           -0.811             0.000
   9 theta   Free           -0.001             0.000
  10   u_x  Fixed            0.000           -60.832
  11   u_y  Fixed            0.000           210.832
  12 theta   Free           -0.027             0.000
```

In [2115…

```
# Element 1, 2, 3
print_element(1, u_global, m1, T1, k1, Qf1)
print_element(2, u_global, m2, T2, k2, Qf2)
print_element(3, u_global, m3, T3, k3, Qf3)
```

```
Element 1
------------------------------------------------------------
u (global) [m&rad] : [ 0.     0.     0.     0.092 -0.    -0.007]
v (local)  [m&rad] : [ 0.     0.     0.    -0.    -0.092 -0.007]
q (local)  [kN&kN·m]    : [ 89.168 125.918 389.592 -89.168 -29.9
18   0.   ]

Element 2
------------------------------------------------------------
u (global) [m&rad] : [ 0.092 -0.    -0.007  0.091 -0.001 -0.001]
v (local)  [m&rad] : [ 0.092 -0.    -0.007  0.091 -0.001 -0.001]
q (local)  [kN&kN·m]    : [  60.832   89.168    0.     -60.832
210.832 -304.158]

Element 3
------------------------------------------------------------
u (global) [m&rad] : [ 0.     0.    -0.027  0.091 -0.001 -0.001]
v (local)  [m&rad] : [ 0.     0.    -0.027 -0.001 -0.091 -0.001]
q (local)  [kN&kN·m]    : [ 210.832   60.832   -0.    -210.832
-60.832  304.158]
```

PART 7 — Wrap-Up

Wrap-Up

In this lecture, we:

- Defined **member releases** and enforced hinge conditions ($M_b = 0$ and/or $M_e = 0$)
- Standardized FEF notation:
  - $F^F_{xb}, F^F_{yb}, F^F_{mb}, F^F_{xe}, F^F_{ye}, F^F_{me}$
- Derived modified member relations for:
  - **MT = 1** (hinge at b)
  - **MT = 2** (hinge at e)
  - **MT = 3** (hinges at both ends)
- Discussed **hinged joints** and why they can make $\mathbf{K}$ singular

Next up:

- Secondary effects (support settlement, temperature, fabrication errors)
- How they enter DSM through equivalent nodal effects / modified $\mathbf{Q}_f$